



4.6 The normal distribution

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A continuous random variable $X \sim N(\mu, \sigma^2)$ has a symmetric, bell-shaped distribution: approximately 68% of values lie within 1σ of the mean, 95% within 2σ , and 99.7% within 3σ . Probabilities are found using technology or the standardised z-score $z = (x - \mu)/\sigma$, referring to the standard normal distribution $N(0, 1)$. The inverse normal function finds the x-value corresponding to a given cumulative probability, useful for problems like finding a cut-off value.

Worked example: follow the reasoning

Adult men's heights in a city are normally distributed with mean 175 cm and standard deviation 7 cm. Find (a) $P(X > 185)$, (b) the height exceeded by only 10% of men.

Solution: (a) $z = (185 - 175)/7 \approx 1.43$; $P(X > 185) = 1 - \Phi(1.43) \approx 1 - 0.9236 = 0.0764$. (b) Need $P(X > x) = 0.10$, so $P(X \leq x) = 0.90$, $z = \text{invNorm}(0.90) = 1.2816$; $x = 175 + 1.2816(7) \approx 184.0$ cm.

Quick check · multiple choice

$X \sim N(50, 16)$ (variance 16). What is the standard deviation?

- A 4
- B 16
- C 8
- D 2

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The weights of bags of flour produced by a machine are normally distributed with mean 1000 g and standard deviation 12 g. (a) Find the probability a randomly selected bag weighs between 990 g and 1015 g, to 3 significant figures. (b) The machine is adjusted so only 2% of bags weigh less than a new minimum acceptable weight. Find this minimum weight, to 1 decimal place. (c) Given $P(X > 1015) \approx 0.106$ from part (a)'s working, find the probability that exactly 2 out of 5 randomly selected bags weigh more than 1015 g. [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $z_1 = (990 - 1000)/12 = -0.833$, $z_2 = (1015 - 1000)/12 = 1.25$;
 $P(990 < X < 1015) = \Phi(1.25) - \Phi(-0.833) \approx 0.8944 - 0.2023 = 0.692$. (b) $P(X < x) = 0.02 \rightarrow$
 $z = \text{invNorm}(0.02) \approx -2.054$; $x = 1000 - 2.054(12) \approx 975.4$ g. (c) $P(X > 1015) = 1 - 0.8944 \approx 0.1056$;
with $Y \sim B(5, 0.1056)$, $P(Y = 2) = C(5, 2)(0.1056)^2(0.8944)^3 \approx 10 \times 0.01115 \times 0.7155 \approx 0.0798$.

Quick check answer

A. $\sigma = \sqrt{16} = 4$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.